Minh-Nhat Nguyen

Undergraduate AI Researcher

Ho Chi Minh City, Vietnam nguyenminhnhat.cs@gmail.com xdnhatnguyen

Summary

A Computer Science undergraduate with a solid mathematical foundation and hands-on research in Computer Vision and Generative Models (Diffusion Models, Flow Matching). Experienced in developing training-free generative methods, multimodal video retrieval, and architectural document understanding pipelines.

Education

Bachelor of Information Technology — **University of Science, VNUHCM**, Information Technology (Computer Science) — Ho Chi Minh City, Vietnam, 2025 – 2029
- GPA: 3.74/4.0 (8.75/10.0), All core mathematics and CS courses: 4.0/4.0.
- Expected graduation: 2029
- Advanced Online Coursework: MIT 6.S184: Generative AI with Stochastic Differential Equations.

High School Diploma — **Le Quy Don Gifted High School**, Mathematics — Da Nang, Vietnam, 2022 – 2025
- Specialized in Advanced Mathematics.

Experience

Autonomous AI Lab, AI Research Intern — Ho Chi Minh City, Vietnam, Oct 2026 – present, 2 months
- Engineered an automated architectural drawing recognition and component counting system from vector PDF floorplans, accurately detecting doors and structural building elements.
- Developed preprocessing and geometric post-processing algorithms integrated with Vision Foundation Models (VFMs) to significantly improve recognition precision.
- Implemented semi-supervised fine-tuning tailored to heterogeneous CAD datasets and deployed the complete inference recognition pipeline.

University of Science, VNUHCM, Undergraduate AI Researcher — Ho Chi Minh City, Vietnam, Mar 2026 – present, 9 months
- Developed training-free methods for style transfer using Latent Diffusion Models and Flow Matching, introducing latent patch-shuffling to eliminate content leakage and pixel-space grid artifacts.
- Optimized generative algorithms by manipulating sampling trajectories, guidance terms, and cross-attention maps to balance stylistic and structural fidelity.

Publications

OrthoStyle: Manifold-Preserving Training-Free Style Transfer via Score-Orthogonal Guidance — Oct 2026

Proposed a training-free framework balancing style fidelity and content preservation without network fine-tuning or DDIM inversion. Demonstrates superior structural consistency and completely outperforms baselines using the same Stable Cascade backbone.

Minh Nhat Nguyen, Dang Quang Pham, Thien Phuc Vo, Phuc Loc Dang, Duc Khoan Le

xdnhatnguyen.github.io/OrthoStyle (The 15th International Symposium on Information and Communication Technology (SOICT 2026) - Oral Presentation)

Awards

Quarter-finalist (Top 36 / 200+ teams) Ho Chi Minh City, Vietnam
Coding Challenge 2026, Faculty of Information Technology, University of Science, VNUHCM May 2026

Projects

Video Retrieval Pipeline — AI Challenge HCMC 2026 Aug 2026
Team lead building an end-to-end multimodal video retrieval pipeline addressing KIS, VQA, and temporal alignment using LanceDB hybrid search (SigLIP + BM25) and dynamic programming.

Latent Patch-Shuffle Style Transfer Aug 2026
Researched and implemented a training-free style transfer framework on latent diffusion models.

Handwritten Digits Classification Nov 2025
Led research benchmarking CNN vs. MLP architectures from scratch in PyTorch for the MNIST dataset.